Haodong Zou

Embodied AI · Robot Learning · Dexterous Manipulation · Sim-to-Real
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Research Interests

Learning transferable skills for humanoid locomotion, whole-body motion tracking, loco-manipulation, and dual-arm dexterous manipulation from human videos and egocentric demonstrations. Focused on motion retargeting, reinforcement and imitation learning, generalization across terrains and object interactions, and sim-to-real transfer.

Education

National University of Singapore Aug. 2026 – Present
M.Sc. in Robotics, expected Jul. 2027
LV Robotics Lab (Prof. Shuicheng Yan); Grasping and Manipulation Lab (Prof. Jianshu Zhou).
Coursework: Machine Learning, Robotics, Computer Vision.

Beijing Institute of Technology Sep. 2022 – Jul. 2026
B.Eng. in Automation *GPA 88.5/100*
Second-Class and Third-Class Scholarships; Outstanding Student Award.

Research and Internship Experience

Research Student, LV Robotics Lab & G&M Lab, NUS Aug. 2026 – Present
Egocentric data and transfer for long-horizon kitchen contact tasks

- Built an end-to-end pipeline from egocentric human recordings to dual-arm dexterous execution in simulation; now refining pose accuracy and retargeting quality.
- Recovering camera pose from RGB, metric depth, and head pose; fitting optical-mocap skeletons to bimanual MANO; estimating world-frame 6-DoF poses of multiple rigid objects with SAM 3 / SAM 3D and PCA/ICP.
- Retargeting human hand demonstrations to the Wuji Hand with mink inverse kinematics, then refining the motion with MPPI in MuJoCo.
- Placing free-wrist trajectories on a dual Franka FR3 and solving only the 14 arm joints for deployment.

RL Algorithm Intern, OakCreek Tech, Beijing Mar. 2026 – Jun. 2026
Humanoid general-purpose software; video-to-robot motion (GVHMR, GMR, NMR / BeyondMimic)

- **Locomotion (velocity tracking):** improved blind and perception-based locomotion within an existing teacher–student framework and deployed the policies on hardware. Set command headings from nearby flat-ground samples, added terrain labels to policy inputs, and penalized velocity direction errors.
- **Motion tracking (video-to-dance pipeline):** built a GVHMR-GMR / NMR-BeyondMimic pipeline; refined motor power and torque modeling and added terrain and center-of-mass randomization for sim-to-real robustness.
- **Motion data cleaning:** built a keyframe editor and trajectory post-processor; compared dance-specific smoothing, HMR retargeting to G1 followed by GMR transfer to other robots, and physics filtering via RL motion tracking in simulation. HMR performed best, followed by physics filtering.
- **Loco-manipulation:** reproduced HDMI, OmniRetarget, and SONIC. HDMI and OmniRetarget worked in simulation but generalized poorly on hardware to carrying different-sized boxes and climbing boxes of varied heights. Replaced motion capture in HDMI data collection with low-cost AprilTag tracking.

Undergraduate Thesis, Beijing Institute of Technology Dec. 2025 – Apr. 2026
Teacher–student reinforcement learning for Unitree G1 locomotion

- Trained a privileged PPO teacher in Isaac Gym with a terrain curriculum; distilled it with DAgger into a student that sees only proprioceptive history; aligned the observation gap with WGAN-GP.
- Raised the traversal rate on a fixed pyramid terrain in simulation from 46.7% to 68.3%.

Selected Honors

- Provincial First Prize (Physics Experiment Competition) and Third Prize (Physics Tournament), 2023.
- Second Prize, "Century Cup" Student Academic and Technology Competition, BIT (2025).

Skills

Programming: Python, C/C++, PyTorch, CUDA; MATLAB/Simulink, STM32. **English:** IELTS 7.0.
Robotics: Isaac Gym, Isaac Lab, MuJoCo, mjlab, ROS; reinforcement learning, imitation learning, motion retargeting, humanoid locomotion and loco-manipulation, sim-to-real.